**Faculty of Computer Science and Engineering**
**Ghulam Ishaq Khan Institute of Engineering Sciences and Technology**

Instructor: **Dr. Ali Imran Sandhu**
Undergraduate Course: **Artificial Intelligence**
**Due Date**: 12PM, Oct 14, 2026

Assignment Number: **2**
Date: October 10, 2026
Mapping: **CLO 1 - GA 2 - C2**

## Problem 1: Logistic Regression (Binary Only)

A bank uses logistic regression ($\theta \in 4.2, -2.5$) to predict loan default based on debt-to-income ratio, where $x$ = debt-to-income ratio, $p$ = probability of default.

**Part A:** Calculate the probability of default for $x = 0.3, 0.6, 0.8$

**Part B:** Using threshold 0.5, classify each customer.

**Part C:** Should the bank decrease threshold to 0.3 or increase to 0.7 for fewer false negatives? Explain.

## Problem 2: Medical Diagnosis - Confusion Matrix

A diagnostic test for COVID-19 was administered to 1,000 patients. The results are summarized below:

- 100 patients actually have COVID-19
- 900 patients do not have COVID-19
- The test correctly identified 85 patients with COVID-19
- The test incorrectly identified 45 healthy patients as having COVID-19

**Part A:** Construct the confusion matrix with proper labels.

**Part B:** Calculate:

(a) **Accuracy:**

(b) **Precision:**

(c) **Recall (Sensitivity):**

(d) **Specificity:**

**Part C:** In a medical context, which metric is more important – precision or recall? Justify your answer considering the consequences of false positives vs false negatives in COVID-19 testing.

## Problem 3: Threshold Tuning

Consider two different applications using the same binary classification model but requiring different optimization strategies.

**Application A: Email Spam Detection**
Current Performance at threshold = 0.5:

- Precision = 0.92, Recall = 0.78, Accuracy = 0.89

**Application B: Cancer Screening**
Current Performance at threshold = 0.5:

- Precision = 0.85, Recall = 0.72, Accuracy = 0.88

**Part A:** For each application, determine whether you would:

(a) Increase the threshold (be more conservative)

(b) Decrease the threshold (be more liberal)

(c) Keep the current threshold

Provide detailed reasoning for each decision.

**Part B:** Explain the business/medical implications of false positives and false negatives for each application.

**Part C:** If you decrease the threshold from 0.5 to 0.3 for the cancer screening application:

(a) How would you expect precision to change?

(b) How would you expect recall to change?

(c) Why is this trade-off acceptable in medical screening?